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جامعة الأقصى
كلية الحاسبات وتكنولوجيا المعلومات
قسم علم الحاسوب والمعلومات

Advanced Algorithm

Answers Of Assignment 3

Gaussian Elimination

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Q1/ Solve using Gaussian Elimination:

→ Advanced Algorithm
→ Assignment "3"

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[1] Solve Using Gaussian Elimination :

$$2x + 3y - z = 5$$
$$4x + 4y - 3z = 3$$
$$2x + 3y + 2z = 7$$

①
$$\begin{array}{cccc|c} & x & y & z & \text{الثوابت} \\ R_1 & 2 & 3 & -1 & 5 \\ R_2 & 4 & 4 & -3 & 3 \\ R_3 & 2 & 3 & 2 & 7 \end{array}$$

② Forward Elimination :

$R_2 = R_2 - \frac{4}{2}R_1 \Rightarrow \boxed{R_2 - 2R_1}$

$R_3 = R_3 - \frac{2}{2}R_1 \Rightarrow \boxed{R_3 - R_1}$

$$\left[\begin{array}{ccc|c} 2 & 3 & -1 & 5 \\ 0 & -2 & -1 & -7 \\ 0 & 0 & 3 & 2 \end{array}\right]$$ ✓ Upper trangular matrix

③ Back substitution :

$3z = 2 \Rightarrow \boxed{z = \frac{2}{3}}$ ※

$-2y - z = -7 \rightarrow -2y - \frac{2}{3} = -7 \quad -2y = -7 + \frac{2}{3} \Rightarrow \boxed{y = \frac{19}{6}}$ ※

$2x + 3y - z = 5 \rightarrow 2x + 3 \times \frac{19}{6} - \frac{2}{3} = 5 \Rightarrow \boxed{x = \frac{4}{3}}$ ※

Q2/ Pseudo code of Gaussian Elimination:

[2] Pseudo code for Gaussian Elimination with Pivoting :

```
Input: Augmented matrix A of size n x (n+1)
for i = 1 to n-1 do
        // Pivoting : find row with max absolute value in column i from rows i to n.
      max row = i
      for k = i+1 to n do
          if abs (A [k, i ]) > abs (A [ max Row, i ]) then
                    max row = k
      if max Row != i then
              swap rows A[i] and A [max Row]

        // Forward elimination
      for j = i+1 to n do
          factor = A [j, i ] / A [ i, i ]
          for k = i to n+1 do
              A [ j, k ] = A [ j, k ] - factor * A [ i, k ]

      // Back substitution
      x = array of size n
      for i = n downto 1 do
              sum = 0
              for j = i+1 to n do
                  sum = sum + A [i, j ] * x [j]
              x [i] = (A [ i, n+1 ] - sum) / A[ i, i ]

Output : solution vector X
```

→ Pseudocode : حل آخر

```
Input : Augmented matrix A[n][n+1]
for i = 1 to n-1 :
    for j = i+1 to n:
        factor = A[j][i] / A[i][i]
        for k = i to n+1 :
            A[j][k] = A[j][k] - factor * A[i][k].

Back Substituion :
for i = n downto 1 :
    sum = 0
    for j = i+1 to n :
        sum += A[i][j] * x[j]
    x[i] = (A[i][n+1] - sum) / A[i][i]
```

Q3/ The Implementation of Gaussian Elimination In Java Code:

```java
package TransformAndConqer;
// Deema Mohammed AL-Maqadma
public class GaussianElimination {

    public static void main(String[] args) {
        int n = 3;
        double[][] a = {
            {2, 3, -1, 5},
            {4, 4, -3, 3},
            {2, 3, 2, 7}
        };
        // Forward Elimination with Pivoting
        for (int i = 0; i < n; i++) {
            // Pivoting
            int maxRow = i;
            for (int k = i + 1; k < n; k++) {
                if (Math.abs(a[k][i]) > Math.abs(a[maxRow][i])) {
                    maxRow = k;
                }
            }
            // Swap rows
            double[] temp = a[i];
            a[i] = a[maxRow];
            a[maxRow] = temp;
            // Elimination
            for (int j = i + 1; j < n; j++) {
                double factor = a[j][i] / a[i][i];
                for (int k = i; k <= n; k++) {
                    a[j][k] -= factor * a[i][k];
                }
            }
        }
        // Back Substitution
        double[] x = new double[n];
        for (int i = n - 1; i >= 0; i--) {
            x[i] = a[i][n];
            for (int j = i + 1; j < n; j++) {
                x[i] -= a[i][j] * x[j];
            }
            x[i] /= a[i][i];
        }
        // Output
        System.out.println("Solution:");
        for (int i = 0; i < n; i++) {
            System.out.printf("x%d = %.2f\n", i + 1, x[i]);
        }
```

" واخر دعواهم ان الحمد لله رب العالمين "